

# 011 — 14 Sheet Matriks Implementasi OBE

Dokumen Kurikulum OBE Definitif Program Studi Sistem dan Teknologi Informasi (S1)

## IMPLEMENTASI MODUL OBE — S1 SISTEM DAN TEKNOLOGI INFORMASI UWG 2026

### Program Studi Sistem dan Teknologi Informasi (SISTEKIN)

### Fakultas Sains dan Teknologi Informasi (FSTI) — Universitas Widyagama Malang

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**Standar:** Permendikbudristek No. 53/2023 | APTIKOM OBE IS2020/IT2017 | LAM INFOKOM

**Total Sheet:** 14

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### Sheet 1: Cek List Kesiapan OBE

*SELF ASSESSMENT KESIAPAN IMPLEMENTASI MODUL OBE - S1 SISTEM DAN TEKNOLOGI INFORMASI 2026*

| NO | ASPEK         | KRITERIA / PERTANYAAN KESIAPAN                                      | STATUS | CATATAN                                                                            |
|----|---------------|---------------------------------------------------------------------|--------|------------------------------------------------------------------------------------|
| 1  | Kurikulum OBE | Profil lulusan tersedia dan menggambarkan peran lulusan.            | Ya     | 4 Profil Lulusan (PL-1 s.d. PL-4) telah dirumuskan (Dok. 002).                     |
| 2  | Kurikulum OBE | Jumlah profil lulusan $\geq 3$ dan mencakup bekerja/wirausaha.      | Ya     | 4 PL, termasuk Digital Technopreneur (PL-4).                                       |
| 3  | Kurikulum OBE | Profil lulusan disajikan dengan kode, nama, deskripsi, dan profesi. | Ya     | Tersedia pada Sheet 2 dan Dok. 002.                                                |
| 4  | Kurikulum OBE | CPL prodi tersedia dan dirumuskan dengan KKO + konten + konteks.    | Ya     | 14 CPL (S1, KU1-3, P1-4, KK1-6) pada Dok. 003.                                     |
| 5  | Kurikulum OBE | Jumlah CPL berada pada rentang 10-15.                               | Ya     | Jumlah CPL = 14 (SN-Dikti + APTIKOM IS2020/IT2017).                                |
| 6  | Kurikulum OBE | Semua CPL dipetakan ke Profil Lulusan.                              | Ya     | Tersedia pada Sheet 4 dan Dok. 004.                                                |
| 7  | Kurikulum OBE | Setiap PL ditopang minimal 3 CPL.                                   | Ya     | Seluruh PL ditopang min. 8 CPL.                                                    |
| 8  | Kurikulum OBE | Semua mata kuliah dipetakan terhadap CPL.                           | Ya     | 55 MK (paket) dan 67 MK (portofolio) dipetakan pada Sheet 5 (Dok. 004).            |
| 9  | Kurikulum OBE | Struktur mata kuliah per semester dan total SKS tersedia.           | Ya     | 55 MK / 146 SKS paket ditempuh; 67 MK / 182 SKS portofolio; 8 semester (Dok. 005). |
| 10 | Kurikulum OBE | Peta pemenuhan CPL oleh mata kuliah (level I-                       | Ya     | Tersedia pada Sheet 7 (Dok. 004).                                                  |

| NO | ASPEK         | KRITERIA / PERTANYAAN KESIAPAN                                      | STATUS | CATATAN                                                                                                                                          |
|----|---------------|---------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------|
|    |               | R-M) tersedia.                                                      |        |                                                                                                                                                  |
| 11 | Kurikulum OBE | Bahan Kajian (BK) telah difinalkan dan dipetakan CPL-BK-MK.         | Ya     | 21 BK IS2020 + 15 BK IT2017 dipetakan penuh, mencakup 11 BK kompetensi utama IS2020 & 14 BK penciri utama IT2017 yang wajib diadopsi (Dok. 003). |
| 12 | OBLT          | Setiap mata kuliah memiliki RPS berbasis CPL/CPMK/Sub-CPMK.         | Ya     | CPMK dan Sub-CPMK seluruh 67 MK tersedia pada Dok. 007.                                                                                          |
| 13 | OBLT          | Metode pembelajaran selaras dengan CPMK dan asesmen.                | Ya     | Case Method + PjBL $\geq$ 60% sesuai IKU 7; diverifikasi Dok. 008.                                                                               |
| 14 | OBLAE         | Rencana evaluasi dan bobot penilaian setiap MK tersedia.            | Ya     | Skema 4x Asesmen (T1=20%, UTS=25–30%, T2=20–25%, UAS=30%) pada Dok. 007 & 008.                                                                   |
| 15 | OBLAE         | Capaian CPMK dan CPL dihitung dari data nilai aktual mahasiswa.     | Belum  | Memerlukan data hasil asesmen aktual implementasi kurikulum baru.                                                                                |
| 16 | OBLAE         | Analisis ketercapaian dan continuous improvement dilakukan berkala. | Belum  | Siklus PPEPP/CQI dirancang pada Dok. 010; dijalankan setelah implementasi.                                                                       |

| NO | ASPEK   | KRITERIA / PERTANYAAN KESIAPAN                                                         | STATUS | CATATAN                                                             |
|----|---------|----------------------------------------------------------------------------------------|--------|---------------------------------------------------------------------|
| 17 | Dokumen | Landasan, tracer study, PEO/Tujuan Pendidikan, dan referensi kurikulum terdokumentasi. | Ya     | VMTS 2045, SWOT, 4 PL + 3 PEO terdokumentasi penuh (Dok. 001, 002). |

## Sheet 2: Profil Lulusan (4 PL)

### PROFIL LULUSAN - S1 SISTEM DAN TEKNOLOGI INFORMASI 2026 (Dok. 002)

| NO | KODE PL | NAMA PROFIL LULUSAN                                            | DESKRIPSI PROFIL LULUSAN                                                                                                                                                                                             | PROFESI / BIDANG KARIER                                                                                                 |
|----|---------|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 1  | PL-1    | Intelligent IS Developer & AI Engineer                         | Lulusan mampu merancang dan mengembangkan sistem informasi berbasis AI yang terintegrasi, andal, dan adaptif sesuai kebutuhan bisnis dan organisasi, mencakup rekayasa data, model ML, dan integrasi layanan cerdas. | AI Application Developer; Intelligent Enterprise Developer; Software Engineer (AI/ML); Data Scientist; MLOps Specialist |
| 2  | PL-2    | Cloud Infrastructure, Cybersecurity & Smart Systems Integrator | Lulusan mampu mengintegrasikan infrastruktur cloud, IoT, dan keamanan informasi menjadi sistem cerdas yang aman, andal, dan sesuai standar GRC (COBIT/ITIL/ISO 27001).                                               | Cloud Systems Integrator; IoT Solutions Architect; Smart City Engineer; IT Auditor; Cybersecurity GRC Specialist        |

| NO | KODE PL | NAMA PROFIL LULUSAN                          | DESKRIPSI PROFIL LULUSAN                                                                                                                                                                            | PROFESI / BIDANG KARIER                                                                           |
|----|---------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 3  | PL-3    | UI/UX Designer & Digital Platform Engineer   | Lulusan mampu merancang pengalaman pengguna (UX) yang inklusif dan membangun platform digital skalabel berbasis web, mobile, microservices, dan proses bisnis terotomasi.                           | Full-Stack Platform Engineer; UX/Product Engineer; Mobile Lead Developer; Digital Product Manager |
| 4  | PL-4    | Digital Technopreneur & IT Product Innovator | Lulusan mampu mengembangkan usaha dan inovasi digital berbasis kebutuhan pasar dan masyarakat secara kreatif, etis, dan berkelanjutan, termasuk memimpin tim dan mengelola proyek TI skala startup. | Tech Startup Founder; SaaS Product Owner; Digital Product Manager; IT Project Manager             |

## Sheet 3: CPL Prodi (14 CPL)

*CAPAIAN PEMBELAJARAN LULUSAN (CPL) - 14 OUTCOMES (S1 + KU1-3 + P1-4 + KK1-6) CPL dirumuskan sesuai SN-Dikti + APTIKOM IS2020/IT2017. Sumber: Dok. 003.*

| NO | KODE CPL | KATEGORI | RUMUSAN CPL                                                                    | FOKUS / DOMAIN            | PL TERK.            |
|----|----------|----------|--------------------------------------------------------------------------------|---------------------------|---------------------|
| 1  | S1       | Sikap    | Bertakwa kepada Tuhan Yang Maha Esa, menginternalisasi nilai, norma, dan etika | Sikap & etika profesional | PL-1<br>s.d. P<br>4 |

| NO | KODE CPL | KATEGORI          | RUMUSAN CPL                                                                                                                                                                                                   | FOKUS / DOMAIN                      | PL TERK    |
|----|----------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------|
|    |          |                   | digital, menjunjung tinggi kejujuran dan profesionalisme, bekerja sama secara produktif dalam tim lintas disiplin, serta menunjukkan sikap tanggung jawab atas pekerjaan di bidang keahliannya secara mandiri |                                     |            |
| 2  | KU1      | Keterampilan Umum | Mampu menerapkan pemikiran logis, kritis, sistematis, dan inovatif dalam menganalisis permasalahan computing serta mengkaji implikasi pengembangan teknologi sesuai kaidah ilmiah                             | Analisis & inovasi computing        | PL-1, PL-2 |
| 3  | KU2      | Keterampilan Umum | Mampu mengkomunikasikan gagasan, analisis, dan solusi teknis secara efektif secara lisan dan tulisan, serta memelihara dan mengembangkan jaringan kerja profesional                                           | Komunikasi & networking profesional | PL-3, PL-4 |

| NO | KODE CPL | KATEGORI          | RUMUSAN CPL                                                                                                                                                                                      | FOKUS / DOMAIN                         | PL TERK.   |
|----|----------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|------------|
| 4  | KU3      | Keterampilan Umum | Mampu mengambil keputusan tepat berdasarkan analisis informasi dan data, bertanggungjawab atas hasil kerja kelompok, melakukan evaluasi diri, serta mengelola pembelajaran secara mandiri        | Pengambilan keputusan & manajemen diri | PL-3, PL-4 |
| 5  | P1       | Pengetahuan       | Mempunyai pengetahuan dasar sains dan matematika (kalkulus, statistika, aljabar linear, matematika diskrit, logika) sebagai fondasi ilmu komputasi dan pemecahan masalah computing yang kompleks | Fondasi sains & matematika komputasi   | PL-1, PL-2 |
| 6  | P2       | Pengetahuan       | Memahami konsep dasar sistem informasi, metodologi pengembangan sistem, analisis dan desain sistem, serta peran SI dalam mengelola data dan memberikan rekomendasi keputusan organisasi          | Sistem informasi & metodologi SI       | PL-1, PL-3 |

| NO | KODE CPL | KATEGORI                 | RUMUSAN CPL                                                                                                                                              | FOKUS / DOMAIN                           | PL TERK.   |
|----|----------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|------------|
| 7  | P3       | Pengetahuan              | Memahami konsep infrastruktur TI, arsitektur jaringan, cloud, keamanan informasi, tata kelola TI, etika digital, dan prinsip manajemen proyek            | Infrastruktur TI & keamanan              | PL-2       |
| 8  | P4       | Pengetahuan              | Memahami konsep pengelolaan data, basis data, analitik data, visualisasi data, pemrograman web dan mobile, serta UI/UX untuk pengembangan solusi digital | Platform digital & pengembangan aplikasi | PL-1, PL-3 |
| 9  | KK1      | Keterampilan Khusus (P1) | Mampu menganalisis, merancang, dan mengembangkan sistem informasi cerdas berbasis AI/ML untuk solusi bisnis dan organisasi                               | AI/ML integration ke SI bisnis           | PL-1       |
| 10 | KK2      | Keterampilan Khusus (P1) | Mampu mengumpulkan, memproses, dan menganalisis data serta membangun model machine learning untuk solusi bisnis (data                                    | Data engineering & ML pipeline           | PL-1       |



| NO | KODE CPL | KATEGORI                 | RUMUSAN CPL                                                                                                                                                   | FOKUS / DOMAIN                          | PL TERK. |
|----|----------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|----------|
|    |          |                          | engineering, BI, MLOps)                                                                                                                                       |                                         |          |
| 11 | KK3      | Keterampilan Khusus (P2) | Mampu mengintegrasikan dan mengelola infrastruktur cloud, menerapkan keamanan siber, serta menjalankan praktik DevOps (CI/CD, containerization, IaC)          | Cloud, IoT & infrastruktur terintegrasi | PL-2     |
| 12 | KK4      | Keterampilan Khusus (P2) | Mampu menerapkan tata kelola TI, melakukan audit keamanan, dan memastikan kepatuhan terhadap regulasi dan standar industri (COBIT, ITIL, ISO)                 | Audit, GRC & tata kelola TI             | PL-2     |
| 13 | KK5      | Keterampilan Khusus (P3) | Mampu merancang dan mengimplementasikan antarmuka pengguna (UX/UI) yang efektif serta platform digital yang skalabel (microservices, SaaS, cloud-native, API) | Platform digital skalabel & UI/UX       | PL-3     |
| 14 | KK6      | Keterampilan Khusus (P3) | Mampu merancang dan mengelola proyek startup teknologi informasi cerdas                                                                                       | Manajemen proyek & technopreneurship    | PL-4     |

| NO | KODE CPL | KATEGORI | RUMUSAN CPL                                                                              | FOKUS / DOMAIN | PL TERK |
|----|----------|----------|------------------------------------------------------------------------------------------|----------------|---------|
|    |          |          | dengan model bisnis berkelanjutan (lean startup, MVP, scaling, digital entrepreneurship) |                |         |

## Sheet 4: Matriks CPL – Profil Lulusan

*MATRIKS KETERKAITAN 14 CPL - 4 PROFIL LULUSAN (Dok. 004)*

| NO | KODE CPL | RUMUSAN CPL (RINGKAS)                         | PL-1 | PL-2 | PL-3 | PL-4 | JUMLAH PL |
|----|----------|-----------------------------------------------|------|------|------|------|-----------|
| 1  | S1       | Sikap & etika profesional digital             | V    | V    | V    | V    | 4         |
| 2  | KU1      | Pemikiran logis, kritis & inovatif computing  | V    | V    | V    | V    | 4         |
| 3  | KU2      | Komunikasi efektif lisan/tulisan & networking | V    | V    | V    | V    | 4         |
| 4  | KU3      | Pengambilan keputusan & manajemen diri        | V    | V    | V    | V    | 4         |
| 5  | P1       | Fondasi sains & matematika komputasi          | V    | V    | V    | V    | 4         |

| NO                | KODE CPL | RUMUSAN CPL (RINGKAS)                    | PL-1 | PL-2 | PL-3 | PL-4 | JUMLAH PL                           |
|-------------------|----------|------------------------------------------|------|------|------|------|-------------------------------------|
| 6                 | P2       | Sistem informasi & metodologi SI         | V    |      | V    |      | 2                                   |
| 7                 | P3       | Infrastruktur TI & keamanan              |      | V    |      |      | 1                                   |
| 8                 | P4       | Platform digital & pengembangan aplikasi | V    |      | V    | V    | 3                                   |
| 9                 | KK1      | AI/ML integration ke SI bisnis           | V    |      |      |      | 1                                   |
| 10                | KK2      | Data engineering & ML pipeline           | V    |      |      |      | 1                                   |
| 11                | KK3      | Cloud, IoT & infrastruktur terintegrasi  |      | V    |      |      | 1                                   |
| 12                | KK4      | Audit, GRC & tata kelola TI              |      | V    |      |      | 1                                   |
| 13                | KK5      | Platform digital skalabel & UI/UX        |      |      | V    |      | 1                                   |
| 14                | KK6      | Manajemen proyek & technopreneurship     |      |      |      | V    | 1                                   |
| Jumlah CPL per PL |          |                                          | 10   | 9    | 9    | 8    | Kriteria: min. 3 CPL/PL - terpenuhi |

## Sheet 5: Matriks CPL – Mata Kuliah (67 MK Portofolio)

*MATRIKS PEMETAAN CPL – MATA KULIAH (55 MK PAKET / 146 SKS – PORTOFOLIO 67 MK / 182 SKS) V = CPL dibebankan dan diases pada MK tersebut. Sumber: Dok. 004.*

| NO                                               | KODE MK | NAMA MATA KULIAH       | SMT | SKS | S1 | KU1 | KU2 |
|--------------------------------------------------|---------|------------------------|-----|-----|----|-----|-----|
| A. MKWU – MATA KULIAH WAJIB UMUM (8 MK / 13 SKS) |         |                        |     |     |    |     |     |
| 1                                                | MKU-101 | Agama I                | 1   | 2   | V  |     |     |
| 2                                                | MKU-102 | Pancasila              | 1   | 2   | V  |     |     |
| 3                                                | MKU-103 | Bahasa Indonesia       | 1   | 2   |    |     | V   |
| 4                                                | MKU-204 | Kewirausahaan I        | 2   | 2   |    |     |     |
| 5                                                | MKU-405 | Kewarganegaraan        | 4   | 2   | V  |     |     |
| 6                                                | MKU-406 | Agama II               | 4   | 0   | V  |     |     |
| 7                                                | MKU-508 | Kewirausahaan II       | 5   | 0   |    |     |     |
| 8                                                | MKU-507 | KPM (Kuliah Pengabdian | 5   | 3   | V  |     | V   |

| NO                                                       | KODE MK | NAMA MATA KULIAH               | SMT | SKS | S1 | KU1 | KU2 |
|----------------------------------------------------------|---------|--------------------------------|-----|-----|----|-----|-----|
|                                                          |         | Kepada Masyarakat)             |     |     |    |     |     |
| B. MK FSTI - MATA KULIAH WAJIB FAKULTAS (13 MK / 36 SKS) |         |                                |     |     |    |     |     |
| 9                                                        | FST-101 | Dasar Teknologi Digital        | 1   | 2   |    |     |     |
| 10                                                       | FST-102 | Algoritma dan Pemrograman (+P) | 1   | 3   |    | V   |     |
| 11                                                       | FST-203 | Struktur Data & Algoritma (+P) | 2   | 3   |    | V   |     |
| 12                                                       | FST-204 | Pengantar AI & Data            | 2   | 2   |    |     |     |
| 13                                                       | FST-205 | Basic English                  | 2   | 2   |    |     | V   |
| 14                                                       | FST-206 | Etika dan Hukum Digital        | 2   | 2   | V  |     | V   |
| 15                                                       | FST-207 | Basis Data (+P)                | 2   | 3   |    |     |     |
| 16                                                       | FST-408 | Probabilitas dan Statistika    | 4   | 3   |    |     |     |
| 17                                                       | FST-610 | Capstone Project FSTI (+P)     | 7   | 3   | V  | V   | V   |

| NO                                                               | KODE MK | NAMA MATA KULIAH                       | SMT | SKS | S1 | KU1 | KU2 |
|------------------------------------------------------------------|---------|----------------------------------------|-----|-----|----|-----|-----|
| 18                                                               | FST-611 | Metodologi Penelitian                  | 6   | 2   |    | V   | V   |
| 19                                                               | FST-612 | Praktik Kerja Lapangan (+P)            | 7   | 3   | V  | V   | V   |
| 20                                                               | FST-613 | Pra-Skripsi / Seminar Proposal         | 7   | 2   |    | V   | V   |
| 21                                                               | FST-714 | Skripsi / Tugas Akhir (+P)             | 8   | 6   | V  | V   | V   |
| C. MK CORE STI - MATA KULIAH INTI PROGRAM STUDI (28 MK / 79 SKS) |         |                                        |     |     |    |     |     |
| 22                                                               | STI-101 | Pengantar Sistem & Teknologi Informasi | 1   | 2   |    |     |     |
| 23                                                               | STI-102 | Kalkulus                               | 1   | 3   |    |     |     |
| 24                                                               | STI-103 | Arsitektur & Organisasi Sistem TI      | 1   | 3   |    | V   |     |
| 25                                                               | STI-204 | Matematika Diskrit dan Logika          | 2   | 3   |    | V   |     |
| 26                                                               | STI-205 | Aljabar Linear                         | 2   | 3   |    |     |     |
| 27                                                               | STI-306 | Analisis & Perancangan                 | 3   | 3   |    | V   |     |

| NO | KODE<br>MK  | NAMA MATA<br>KULIAH                               | SMT | SKS | S1 | KU1 | KU2 |
|----|-------------|---------------------------------------------------|-----|-----|----|-----|-----|
|    |             | Sistem Informasi                                  |     |     |    |     |     |
| 28 | STI-<br>307 | Sistem Cerdas                                     | 3   | 2   |    |     |     |
| 29 | STI-<br>308 | UI/UX Design &<br>Prototyping (+P)                | 3   | 3   |    |     |     |
| 30 | STI-<br>309 | Rekayasa<br>Perangkat Lunak                       | 3   | 3   |    |     |     |
| 31 | STI-<br>310 | Sistem Operasi                                    | 3   | 3   |    |     |     |
| 32 | STI-<br>311 | Web Front End<br>Development (+P)                 | 3   | 3   |    |     |     |
| 33 | STI-<br>312 | Jaringan<br>Komputer (+P)                         | 3   | 3   |    |     |     |
| 34 | STI-<br>413 | Machine Learning<br>(+P)                          | 4   | 3   |    |     |     |
| 35 | STI-<br>414 | Pengantar NLP &<br>Information<br>Retrieval (+P)  | 4   | 2   |    |     |     |
| 36 | STI-<br>415 | Data Warehouse<br>& Business<br>Intelligence (+P) | 4   | 3   |    |     |     |
| 37 | STI-<br>417 | Komputasi Awan<br>(Cloud<br>Computing)            | 4   | 3   |    |     |     |

| NO | KODE MK | NAMA MATA KULIAH                          | SMT | SKS | S1 | KU1 | KU2 |
|----|---------|-------------------------------------------|-----|-----|----|-----|-----|
| 38 | STI-418 | Dasar Keamanan Informasi                  | 4   | 2   |    |     |     |
| 39 | STI-416 | Web Back End Development (+P)             | 4   | 3   |    | V   |     |
| 40 | STI-519 | Keamanan Informasi Lanjut                 | 5   | 3   |    |     |     |
| 41 | STI-520 | Data Mining & Visualisasi Data (+P)       | 5   | 3   |    |     |     |
| 42 | STI-521 | Internet of Things (IoT) (+P)             | 5   | 3   |    |     |     |
| 43 | STI-522 | Pemrograman Aplikasi Mobile (+P)          | 5   | 3   |    |     |     |
| 44 | STI-523 | Manajemen Proyek Teknologi Informasi      | 5   | 3   |    |     |     |
| 45 | STI-624 | Integrasi Layanan Cerdas Berbasis AI (+P) | 6   | 3   |    |     |     |
| 46 | STI-625 | Smart City & Pemerintahan Digital         | 6   | 2   |    |     |     |
| 47 | STI-626 | Deep Learning & Neural Networks (+P)      | 6   | 3   |    |     |     |



| NO                                                              | KODE MK | NAMA MATA KULIAH                               | SMT | SKS | S1 | KU1 | KU2 |
|-----------------------------------------------------------------|---------|------------------------------------------------|-----|-----|----|-----|-----|
| 48                                                              | STI-627 | Digital Platform Engineering (+P)              | 6   | 3   |    |     |     |
| 49                                                              | STI-728 | Inovasi Teknologi & Startup Digital (+P)       | 7   | 3   |    |     |     |
| D. MK PILIHAN P1:<br>INTEGRATED SMART SYSTEMS<br>(6 MK /18 SKS) |         |                                                |     |     |    |     |     |
| 50                                                              | STA-501 | Decision Support Systems (+P)                  | 5-7 | 3   |    |     |     |
| 51                                                              | STA-601 | Computational Methods & Numerics (+P)          | 5-7 | 3   |    |     |     |
| 52                                                              | STA-602 | Intelligent Agent Systems (+P)                 | 5-7 | 3   |    |     |     |
| 53                                                              | STA-701 | MLOps and AI Pipeline (+P)                     | 5-7 | 3   |    |     |     |
| 54                                                              | STA-702 | Conversational AI & Intelligent Assistant (+P) | 5-7 | 3   |    |     |     |
| 55                                                              | STA-703 | Smart Surveillance & IoT Analytics (+P)        | 5-7 | 3   |    |     |     |
| E. MK PILIHAN P2:<br>CLOUD INFRASTRUCTURE                       |         |                                                |     |     |    |     |     |

| NO                                                                        | KODE<br>MK  | NAMA MATA<br>KULIAH                             | SMT | SKS | S1 | KU1 | KU2 |
|---------------------------------------------------------------------------|-------------|-------------------------------------------------|-----|-----|----|-----|-----|
| &<br>CYBERSECURITY<br>(6 MK /18 SKS)                                      |             |                                                 |     |     |    |     |     |
| 56                                                                        | STB-<br>501 | Network Security<br>& Digital<br>Forensics (+P) | 5-7 | 3   |    |     |     |
| 57                                                                        | STB-<br>601 | Cloud<br>Architecture &<br>DevOps (+P)          | 5-7 | 3   |    |     |     |
| 58                                                                        | STB-<br>602 | Cybersecurity<br>Risk Management                | 5-7 | 3   |    |     |     |
| 59                                                                        | STB-<br>701 | IT Governance &<br>Compliance<br>(COBIT 2019)   | 5-7 | 3   |    |     |     |
| 60                                                                        | STB-<br>702 | IT Service<br>Management (ITIL<br>4)            | 5-7 | 3   |    |     |     |
| 61                                                                        | STB-<br>703 | Enterprise<br>Architecture<br>(TOGAF)           | 5-7 | 3   |    |     |     |
| F. MK PILIHAN P3:<br>DIGITAL<br>PLATFORM<br>ENGINEERING (6<br>MK /18 SKS) |             |                                                 |     |     |    |     |     |
| 62                                                                        | STC-<br>501 | User Experience<br>Research &<br>Design (+P)    | 5-7 | 3   |    |     |     |

| NO                           | KODE MK | NAMA MATA KULIAH                         | SMT | SKS | S1 | KU1 | KU2 |
|------------------------------|---------|------------------------------------------|-----|-----|----|-----|-----|
| 63                           | STC-601 | Rekayasa & Otomasi Proses Bisnis (+P)    | 5-7 | 3   |    |     |     |
| 64                           | STC-602 | Rekayasa Aplikasi Industri Vertikal (+P) | 5-7 | 3   |    |     |     |
| 65                           | STC-701 | Immersive Media & XR Development (+P)    | 5-7 | 3   |    |     |     |
| 66                           | STC-702 | SaaS Architecture & Multi-Tenancy (+P)   | 5-7 | 3   |    |     |     |
| 67                           | STC-703 | Digital Product Management & Agile       | 5-7 | 3   |    |     |     |
| Jumlah MK Pendukung tiap CPL |         |                                          |     |     | 9  | 12  | 11  |

## Sheet 6: Organisasi Mata Kuliah Per Semester

ORGANISASI MATA KULIAH - KURIKULUM SISTEKIN 2026 (Dok. 005) Paket Ditempuh:  
55 MK / 146 SKS / Portofolio Ditawarkan: 67 MK / 182 SKS

| SEMESTER | JUMLAH<br>MK | TOTAL<br>SKS | SKS<br>KUMULATIF | DAFTAR MATA<br>KULIAH (SKS)                                                                                                                                                                                                                                                                                               | CPL<br>DOMINAN                          | CATATAN                                     |
|----------|--------------|--------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------------------|
| 1        | 8 MK         | 19<br>SKS    | 19 SKS           | MKU-101<br>Agama I (2) /<br>MKU-102<br>Pancasila (2) /<br>MKU-103<br>Bahasa<br>Indonesia (2) /<br>FST-101 Dasar<br>Teknologi<br>Digital (2) /<br>FST-102<br>Algoritma &<br>Pemrograman<br>(3) / STI-101<br>Pengantar SI &<br>TI (2) / STI-102<br>Kalkulus (3) /<br>STI-103<br>Arsitektur &<br>Organisasi<br>Sistem TI (3) | S1, P1, P2,<br>P3, P4,<br>KU1           | Font<br>kom<br>cod<br>arsit<br>harc<br>sain |
| 2        | 8 MK         | 20<br>SKS    | 39 SKS           | STI-204<br>Matematika<br>Diskrit dan<br>Logika (3) /<br>STI-205<br>Aljabar Linear<br>(3) / FST-203<br>Struktur Data &<br>Algoritma (3) /<br>FST-204<br>Pengantar AI &<br>Data (2) / FST-<br>205 Basic<br>English (2) /                                                                                                    | P1, P2,<br>P4, KU1,<br>KU2, KK1,<br>KK6 | Tepa<br>Patu<br>Perr<br>53/2                |

| SEMESTER | JUMLAH<br>MK | TOTAL<br>SKS | SKS<br>KUMULATIF | DAFTAR MATA<br>KULIAH (SKS)                                                                                                                                                                                                | CPL<br>DOMINAN                       | CAT                                       |
|----------|--------------|--------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------------------------------------|
| 3        | 7 MK         | 20<br>SKS    | 59 SKS           | FST-206 Etika<br>& Hukum<br>Digital (2) /<br>FST-207 Basis<br>Data (3) / MKU-<br>204<br>Kewirausahaan<br>I (2)                                                                                                             | P2, P3,<br>P4, KK1,<br>KK3, KK5      | Fond<br>siste<br>jarin<br>dan             |
|          |              |              |                  | STI-306 APSI<br>(3) / STI-307<br>Sistem Cerdas<br>(2) / STI-308<br>UI/UX Design<br>(3) / STI-309<br>RPL (3) / STI-<br>310 Sistem<br>Operasi (3) /<br>STI-311 Web<br>Front End (3) /<br>STI-312<br>Jaringan<br>Komputer (3) |                                      |                                           |
| 4        | 8 MK         | 21<br>SKS    | 80 SKS           | STI-413<br>Machine<br>Learning (3) /<br>STI-414<br>Pengantar NLP<br>& IR (2) / STI-<br>415 DW & BI (3)<br>/ STI-417<br>Cloud (3) / STI-<br>418 Dasar<br>Keamanan (2) /<br>STI-416 Web<br>Back End (3) /                    | P1, P2, P3,<br>KK1, KK2,<br>KK3, KU1 | STI-<br>Teor<br>2 SK<br>MKL<br>SKS<br>UWC |

| SEMESTER | JUMLAH<br>MK     | TOTAL<br>SKS | SKS<br>KUMULATIF | DAFTAR MATA<br>KULIAH (SKS)                                                                                                                                                                                                 | CPL<br>DOMINAN                             | CATATAN                                   |
|----------|------------------|--------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------------|
| 5        | 7 MK +<br>1x0SKS | 21<br>SKS    | 101 SKS          | FST-408<br>Statistika (3) /<br>MKU-405<br>KWN (2) /<br>MKU-406<br>Agama II (0)                                                                                                                                              |                                            |                                           |
|          |                  |              |                  | STI-519<br>Keamanan<br>Lanjut (3) / STI-<br>520 Data<br>Mining (3) /<br>STI-521 IoT (3) /<br>STI-522 Mobile<br>App (3) / STI-<br>523 Manpro TI<br>(3) / MKU-507<br>KPM (3) / MK<br>Pilihan 1 (3) /<br>MKU-508<br>KWU II (0) | KK1, KK2,<br>KK3, KK5,<br>KK6, S1,<br>KU   | 6 MK<br>SKS<br>Pilih<br>MKL<br>SKS<br>UWC |
| 6        | 7 MK             | 19<br>SKS    | 120 SKS          | STI-624<br>Integrasi AI (3)<br>/ STI-625<br>Smart City (2) /<br>STI-626 Deep<br>Learning (3) /<br>STI-627 Digital<br>Platform Eng<br>(3) / FST-611<br>Metopel (2) /<br>MK Pilihan 2 (3)<br>/ MK Pilihan 3<br>(3)            | P2, P3,<br>KK1, KK3,<br>KK4, KK5,<br>KU1-3 | STI-<br>SKS<br>Prak<br>MBK<br>20 S        |

| SEMESTER | JUMLAH MK | TOTAL SKS | SKS KUMULATIF | DAFTAR MATA KULIAH (SKS)                                                                                                                                                              | CPL DOMINAN                     | CATATAN                                       |
|----------|-----------|-----------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------------|
| 7        | 7 MK      | 20 SKS    | 140 SKS       | STI-728<br>Startup Digital<br>(3) / FST-610<br>Capstone (3) /<br>FST-612 PKL<br>(3) / FST-613<br>Pra-Skripsi (2)<br>/ MK Pilihan 4<br>(3) / MK Pilihan<br>5 (3) / MK<br>Pilihan 6 (3) | KK6, KK3,<br>KK4,<br>KU1-3, S1  | 4 MK<br>SKS<br>Pilih.<br>MBK<br>20 S          |
| 8        | 1 MK      | 6 SKS     | 146 SKS       | FST-714 Skripsi<br>/ Tugas Akhir<br>(6)                                                                                                                                               | S1, KU1-3,<br>seluruh P<br>& KK | Skripsi<br>SKS<br>Non<br>ekui<br>SKS          |
| TOTAL    | 55 MK     | 146 SKS   | 146 SKS       | 49 MK Wajib<br>(128 SKS) + 6<br>MK Pilihan<br>Peminatan (18<br>SKS, Sem 5-7)<br>= 55 MK / 146<br>SKS                                                                                  |                                 | Port<br>MK /<br>Syar<br>Min.<br>(Peri<br>53/2 |

## Sheet 7: Peta Pemenuhan CPL (Level I-R-M)

*PETA PEMENUHAN CPL OLEH MATA KULIAH - LEVEL I/R/M (Dok. 004) Legenda: I = Introduced (C2-C3), R = Reinforced (C3-C4, diperkuat & dipraktikkan), M = Mastered (C5-C6, dikuasai & diases portofolio)*

| KODE MK | NAMA MATA KULIAH                  | SKS | SMT | S1 | KU1 | KU2 | KU3 | P1 | P2 |
|---------|-----------------------------------|-----|-----|----|-----|-----|-----|----|----|
| MKU-101 | Agama I                           | 2   | 1   | I  |     |     |     |    |    |
| MKU-102 | Pancasila                         | 2   | 1   | I  |     |     |     |    |    |
| MKU-103 | Bahasa Indonesia                  | 2   | 1   |    |     | I   |     |    |    |
| FST-101 | Dasar Teknologi Digital           | 2   | 1   |    |     |     |     |    | I  |
| FST-102 | Algoritma dan Pemrograman         | 3   | 1   |    | I   |     |     |    |    |
| STI-101 | Pengantar Sistem & TI             | 2   | 1   |    |     |     |     |    | I  |
| STI-102 | Kalkulus                          | 3   | 1   |    |     |     |     | I  |    |
| STI-103 | Arsitektur & Organisasi Sistem TI | 3   | 1   |    | I   |     |     | I  |    |
| MKU-204 | Kewirausahaan I                   | 2   | 2   |    |     |     |     |    |    |
| STI-204 | Matematika Diskrit dan Logika     | 3   | 2   |    | R   |     |     | R  |    |
| STI-205 | Aljabar Linear                    | 3   | 2   |    |     |     |     | R  |    |



| KODE MK | NAMA MATA KULIAH           | SKS | SMT | S1 | KU1 | KU2 | KU3 | P1 | P2 |
|---------|----------------------------|-----|-----|----|-----|-----|-----|----|----|
| FST-203 | Struktur Data & Algoritma  | 3   | 2   |    | R   |     |     |    |    |
| FST-204 | Pengantar AI & Data        | 2   | 2   |    |     |     |     |    |    |
| FST-205 | Basic English              | 2   | 2   |    |     | I   |     |    |    |
| FST-206 | Etika dan Hukum Digital    | 2   | 2   | R  |     | R   |     |    |    |
| FST-207 | Basis Data                 | 3   | 2   |    |     |     |     |    | R  |
| STI-306 | Analisis & Perancangan SI  | 3   | 3   |    | R   |     |     |    | R  |
| STI-307 | Sistem Cerdas              | 2   | 3   |    |     |     |     |    | R  |
| STI-308 | UI/UX Design & Prototyping | 3   | 3   |    |     |     |     |    |    |
| STI-309 | Rekayasa Perangkat Lunak   | 3   | 3   |    |     |     |     |    |    |
| STI-310 | Sistem Operasi             | 3   | 3   |    |     |     |     |    |    |
| STI-311 | Web Front End Development  | 3   | 3   |    |     |     |     |    |    |

| KODE MK | NAMA MATA KULIAH                      | SKS | SMT | S1 | KU1 | KU2 | KU3 | P1 | P2 |
|---------|---------------------------------------|-----|-----|----|-----|-----|-----|----|----|
| STI-312 | Jaringan Komputer                     | 3   | 3   |    |     |     |     |    |    |
| STI-413 | Machine Learning                      | 3   | 4   |    |     |     |     |    | R  |
| STI-415 | Data Warehouse & BI                   | 3   | 4   |    |     |     |     |    | R  |
| STI-417 | Komputasi Awan (Cloud Computing)      | 3   | 4   |    |     |     |     |    |    |
| STI-418 | Dasar Keamanan Informasi              | 2   | 4   |    |     |     |     |    |    |
| STI-416 | Web Back End Development              | 3   | 4   |    | R   |     |     |    |    |
| FST-408 | Probabilitas dan Statistika           | 3   | 4   |    |     |     |     | R  |    |
| STI-414 | Pengantar NLP & Information Retrieval | 2   | 4   |    |     |     |     |    | R  |
| STI-519 | Keamanan Informasi Lanjut             | 3   | 5   |    |     |     |     |    |    |
| STI-520 | Data Mining & Visualisasi Data        | 3   | 5   |    |     |     |     |    | M  |

| KODE MK | NAMA MATA KULIAH                     | SKS | SMT | S1 | KU1 | KU2 | KU3 | P1 | P2 |
|---------|--------------------------------------|-----|-----|----|-----|-----|-----|----|----|
| STI-521 | Internet of Things (IoT)             | 3   | 5   |    |     |     |     |    |    |
| STI-522 | Pemrograman Aplikasi Mobile          | 3   | 5   |    |     |     |     |    |    |
| STI-523 | Manajemen Proyek TI                  | 3   | 5   |    |     |     |     |    |    |
| MKU-507 | KPM / KKN Digital                    | 3   | 5   | M  |     | M   |     |    |    |
| STI-624 | Integrasi Layanan Cerdas Berbasis AI | 3   | 6   |    |     |     |     |    | M  |
| STI-625 | Smart City & Pemerintahan Digital    | 2   | 6   |    |     |     |     |    |    |
| STI-626 | Deep Learning & Neural Networks      | 3   | 6   |    |     |     |     |    |    |
| STI-627 | Digital Platform Engineering         | 3   | 6   |    |     |     |     |    |    |
| FST-611 | Metodologi Penelitian                | 2   | 6   |    | M   | M   | M   |    |    |
| STI-728 | Inovasi Teknologi & Startup Digital  | 3   | 7   |    |     |     |     |    |    |

| KODE MK  | NAMA MATA KULIAH               | SKS | SMT | S1 | KU1 | KU2 | KU3 | P1 | P2 |
|----------|--------------------------------|-----|-----|----|-----|-----|-----|----|----|
| FST-610  | Capstone Project FSTI          | 3   | 7   | M  | M   | M   | M   |    |    |
| FST-612  | Praktik Kerja Lapangan (PKL)   | 3   | 7   | M  | M   | M   | M   |    |    |
| FST-613  | Pra-Skripsi / Seminar Proposal | 2   | 7   |    | M   | M   | M   |    |    |
| FST-714  | Skripsi / Tugas Akhir          | 6   | 8   | M  | M   | M   | M   | M  | M  |
| Jumlah I |                                |     |     | 2  | 2   | 2   | 0   | 2  | 2  |
| Jumlah R |                                |     |     | 1  | 4   | 2   | 0   | 3  | 7  |
| Jumlah M |                                |     |     | 3  | 4   | 4   | 4   | 1  | 3  |

## Sheet 8: Penurunan CPL ke CPMK & Sub-CPMK (6 MK Penciri)

*PENURUNAN CPL -> CPMK -> SUB-CPMK -> EVALUASI - 6 MK PENCIRI SISTEKIN 2026 (Dok. 007)*

| KODE MK | NAMA MATA KULIAH                     | PL UTAMA | CPMK  | RUMUSAN CPMK                                                                          | SUB-CPMK    | RUM                      |
|---------|--------------------------------------|----------|-------|---------------------------------------------------------------------------------------|-------------|--------------------------|
| STI-624 | Integrasi Layanan Cerdas Berbasis AI | PL-1     | CPMK1 | Menganalisis kebutuhan integrasi layanan cerdas dan data pada suatu kasus organisasi. | Sub-CPMK1.1 | Men kebi peni data siste |
| STI-624 | Integrasi Layanan Cerdas Berbasis AI | PL-1     | CPMK1 | Menganalisis kebutuhan integrasi layanan cerdas dan data pada suatu kasus organisasi. | Sub-CPMK1.2 | Men kons dan data        |
| STI-624 | Integrasi Layanan Cerdas Berbasis AI | PL-1     | CPMK2 | Merancang dan membangun prototipe layanan cerdas berbasis AI yang terintegrasi.       | Sub-CPMK2.1 | Men API/ cerc mek (RES   |
| STI-624 | Integrasi Layanan Cerdas Berbasis AI | PL-1     | CPMK3 | Mengevaluasi kualitas, interoperabilitas, dan kinerja solusi cerdas.                  | Sub-CPMK3.1 | Men inter laya men hasi  |
| STI-413 | Machine Learning (+P)                | PL-1     | CPMK1 | Menerapkan alur kerja rekayasa data ML (EDA, Cleaning, Encoding, Scaling).            | Sub-CPMK1.1 | Mela per deni Nurr       |
| STI-413 | Machine Learning                     | PL-1     | CPMK2 | Mengonstruksi model                                                                   | Sub-CPMK2.1 | Men Reg                  |

| KODE MK | NAMA MATA KULIAH        | PL UTAMA | CPMK  | RUMUSAN CPMK                                                                   | SUB-CPMK    | RUMUSAN                  |
|---------|-------------------------|----------|-------|--------------------------------------------------------------------------------|-------------|--------------------------|
|         | (+P)                    |          |       | Supervised & Unsupervised Learning.                                            |             | Tree K-M learn           |
| STI-413 | Machine Learning (+P)   | PL-1     | CPMK3 | Mengevaluasi performa model dan membangun API inferensi (ML Deployment).       | Sub-CPMK3.1 | Seri RES men Fast        |
| STI-413 | Machine Learning (+P)   | PL-1     | CPMK4 | Menerapkan prinsip keandalan, keamanan, dan dokumentasi sistem ML profesional. | Sub-CPMK4.1 | Men carc tekn solu stak  |
| STI-521 | Internet of Things (+P) | PL-2     | CPMK1 | Menganalisis kebutuhan dan arsitektur sistem IoT (sensor-edge-cloud).          | Sub-CPMK1.1 | Men sens prot dan kone   |
| STI-521 | Internet of Things (+P) | PL-2     | CPMK2 | Mengembangkan prototipe IoT yang terhubung dengan layanan cloud.               | Sub-CPMK2.1 | Akui ESP MQ1 plat        |
| STI-521 | Internet of Things (+P) | PL-2     | CPMK3 | Menguji keandalan dan keamanan dasar sistem IoT.                               | Sub-CPMK3.1 | Peng kone keta doki hasi |

| KODE MK | NAMA MATA KULIAH                  | PL UTAMA | CPMK  | RUMUSAN CPMK                                                                        | SUB-CPMK    | RUMUSAN SUB-CPMK                                                  |
|---------|-----------------------------------|----------|-------|-------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------|
| STI-308 | UI/UX Design & Prototyping (+P)   | PL-3     | CPMK1 | Melakukan riset pengguna dan merumuskan problem statement.                          | Sub-CPMK1.1 | Menganalisis kebutuhan pengguna dan merumuskan problem statement. |
| STI-308 | UI/UX Design & Prototyping (+P)   | PL-3     | CPMK2 | Mengembangkan purwarupa interaktif high-fidelity menggunakan Figma.                 | Sub-CPMK2.1 | Ator dan mengkonfigurasi lingkungan pengembangan mikroservis.     |
| STI-308 | UI/UX Design & Prototyping (+P)   | PL-3     | CPMK3 | Mengevaluasi usability prototype dengan pengujian pengguna berbasis metrik.         | Sub-CPMK3.1 | Usability testing dan laporan rekam jejak pengguna.               |
| STI-627 | Digital Platform Engineering (+P) | PL-3     | CPMK1 | Merancang arsitektur microservices dan API gateway untuk platform digital skalabel. | Sub-CPMK1.1 | Implementasi mikroservis dan dokumentasi API Gateway.             |
| STI-627 | Digital Platform Engineering (+P) | PL-3     | CPMK2 | Membangun pipeline CI/CD dan Infrastructure as Code untuk deployment otomatis.      | Sub-CPMK2.1 | CI/CD pipeline dan konfigurasi GitOps Terraform.                  |

| KODE MK        | NAMA MATA KULIAH                  | PL UTAMA         | CPMK             | RUMUSAN CPMK                                                                        | SUB-CPMK         | RUMUSAN                             |
|----------------|-----------------------------------|------------------|------------------|-------------------------------------------------------------------------------------|------------------|-------------------------------------|
| STI-627        | Digital Platform Engineering (+P) | PL-3             | CPMK3            | Mengevaluasi kinerja dan skalabilitas platform digital dengan load testing.         | Sub-CPMK3.1      | Load testing                        |
| STI-519        | Keamanan Informasi Lanjut         | PL-2             | CPMK1            | Menganalisis teknik serangan lanjutan dan mensimulasikan penetration testing dasar. | Sub-CPMK1.1      | Simulasi OWASP dan Metasploit Suite |
| STI-519        | Keamanan Informasi Lanjut         | PL-2             | CPMK2            | Melakukan digital forensics analysis terhadap sistem file dan log jaringan.         | Sub-CPMK2.1      | Analisis dan koreksi kearifan       |
| STI-519        | Keamanan Informasi Lanjut         | PL-2             | CPMK3            | Menyusun kebijakan keamanan informasi organisasi berbasis ISO 27001.                | Sub-CPMK3.1      | IS Policy dan mitigasi stuc         |
| Validasi Bobot | STI-624: 100% OK                  | STI-413: 100% OK | STI-521: 100% OK | STI-308: 100% OK                                                                    | STI-627: 100% OK | STI-                                |



## Sheet 9: Template Evaluasi OBE – STI-624 (Integrasi AI)

TEMPLATE EVALUASI OBE - STI-624 Integrasi Layanan Cerdas Berbasis AI CPL: P2, KK1 | Target Ketercapaian CPL = 75% Skema Asesmen: Tugas 1=20% (Pk.4), UTS=30% (Pk.8), Tugas 2=20% (Pk.12), UAS=30% (Pk.16)

| No              | NIM        | Nama Mahasiswa | Tugas 1 (20%)         | UTS (30%)  | Tugas 2 (20%) | UAS (30%)  | Nilai Akhir | CPMK1 (%)  | CPMK2 (%)         | CPMK3 (%)  | Status CPL |
|-----------------|------------|----------------|-----------------------|------------|---------------|------------|-------------|------------|-------------------|------------|------------|
| 1               | 1000000000 | 1000000000     | 1000000000            | 1000000000 | 1000000000    | 1000000000 | 1000000000  | 1000000000 | 1000000000        | 1000000000 | 1000000000 |
| 2               | 1000000000 | 1000000000     | 1000000000            | 1000000000 | 1000000000    | 1000000000 | 1000000000  | 1000000000 | 1000000000        | 1000000000 | 1000000000 |
| 3               | 1000000000 | 1000000000     | 1000000000            | 1000000000 | 1000000000    | 1000000000 | 1000000000  | 1000000000 | 1000000000        | 1000000000 | 1000000000 |
| 4               | 1000000000 | 1000000000     | 1000000000            | 1000000000 | 1000000000    | 1000000000 | 1000000000  | 1000000000 | 1000000000        | 1000000000 | 1000000000 |
| 5               | 1000000000 | 1000000000     | 1000000000            | 1000000000 | 1000000000    | 1000000000 | 1000000000  | 1000000000 | 1000000000        | 1000000000 | 1000000000 |
| 10              | 1000000000 | 1000000000     | 1000000000            | 1000000000 | 1000000000    | 1000000000 | 1000000000  | 1000000000 | 1000000000        | 1000000000 | 1000000000 |
| 15              | 1000000000 | 1000000000     | 1000000000            | 1000000000 | 1000000000    | 1000000000 | 1000000000  | 1000000000 | 1000000000        | 1000000000 | 1000000000 |
| 20              | 1000000000 | 1000000000     | 1000000000            | 1000000000 | 1000000000    | 1000000000 | 1000000000  | 1000000000 | 1000000000        | 1000000000 | 1000000000 |
| 25              | 1000000000 | 1000000000     | 1000000000            | 1000000000 | 1000000000    | 1000000000 | 1000000000  | 1000000000 | 1000000000        | 1000000000 | 1000000000 |
| 30              | 1000000000 | 1000000000     | 1000000000            | 1000000000 | 1000000000    | 1000000000 | 1000000000  | 1000000000 | 1000000000        | 1000000000 | 1000000000 |
| Ringkasan Kelas |            |                | Rata-rata Nilai Akhir |            | Target CPL    |            | 75%         |            | % Mahasiswa >= 75 |            |            |
| CPMK1-3         |            |                |                       |            |               |            |             |            |                   |            |            |

## Sheet 10: Template Evaluasi OBE – STI-413 (Machine Learning)

TEMPLATE EVALUASI OBE - STI-413 Machine Learning (+P) CPL: P2, KK1, KK2 | Target Ketercapaian CPL = 75% Skema Asesmen: Tugas 1=20% (Pk.4), UTS=25% (Pk.8), Tugas 2=25% (Pk.12), UAS=30% (Pk.16)

| No                | NIM | Nama Mahasiswa | Tugas 1 (20%) | UTS (25%) | Tugas 2 (25%) | UAS (30%) | Nilai Akhir | CPMK1 (%) | CPMK2 (%) | CPMK3 (%) | CPMK4 (%) | Status CPL            |            |     |
|-------------------|-----|----------------|---------------|-----------|---------------|-----------|-------------|-----------|-----------|-----------|-----------|-----------------------|------------|-----|
| 1                 |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 2                 |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 3                 |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 4                 |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 5                 |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 6                 |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 7                 |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 8                 |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 9                 |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 10                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 11                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 12                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 13                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 14                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 15                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 16                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 17                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 18                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 19                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 20                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 21                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 22                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 23                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 24                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 25                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 26                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 27                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 28                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 29                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| 30                |     |                |               |           |               |           |             |           |           |           |           |                       |            |     |
| Ringkasan Kelas   |     |                |               |           |               |           |             |           |           |           |           | Rata-rata Nilai Akhir | Target CPL | 75% |
| % Mahasiswa >= 75 |     |                |               |           |               |           |             |           |           |           |           | CPMK1-4               |            |     |

## Sheet 11: Template Evaluasi OBE – STI-308 (UI/UX)

TEMPLATE EVALUASI OBE - STI-308 UI/UX Design & Prototyping (+P) CPL: KK5 /  
Target Ketercapaian CPL = 75% Skema Asesmen: Tugas 1=20% (Pk.4), UTS=25%  
(Pk.8), Tugas 2=25% (Pk.12), UAS=30% (Pk.16)

| No              | NIM | Nama Mahasiswa | Tugas 1 (20%)         | UTS (25%) | Tugas 2 (25%) | UAS (30%) | Nilai Akhir | CPMK1 (%) | CPMK2 (%)         | CPMK3 (%) | Status CPL |
|-----------------|-----|----------------|-----------------------|-----------|---------------|-----------|-------------|-----------|-------------------|-----------|------------|
| 1               |     |                |                       |           |               |           |             |           |                   |           |            |
| 2               |     |                |                       |           |               |           |             |           |                   |           |            |
| 3               |     |                |                       |           |               |           |             |           |                   |           |            |
| 4               |     |                |                       |           |               |           |             |           |                   |           |            |
| 5               |     |                |                       |           |               |           |             |           |                   |           |            |
| 10              |     |                |                       |           |               |           |             |           |                   |           |            |
| 15              |     |                |                       |           |               |           |             |           |                   |           |            |
| 20              |     |                |                       |           |               |           |             |           |                   |           |            |
| 25              |     |                |                       |           |               |           |             |           |                   |           |            |
| 30              |     |                |                       |           |               |           |             |           |                   |           |            |
| Ringkasan Kelas |     |                | Rata-rata Nilai Akhir |           | Target CPL    |           | 75%         |           | % Mahasiswa >= 75 |           |            |
| CPMK1-3         |     |                |                       |           |               |           |             |           |                   |           |            |

## Sheet 12: Template Evaluasi OBE – STI-519 (Keamanan)

TEMPLATE EVALUASI OBE - STI-519 Keamanan Informasi Lanjut CPL: P3, KK3, KK4 /  
Target Ketercapaian CPL = 75% Skema Asesmen: Tugas 1=20% (Pk.4), UTS=30%  
(Pk.8), Tugas 2=20% (Pk.12), UAS=30% (Pk.16)

| No              | NIM | Nama Mahasiswa | Tugas 1 (20%)         | UTS (30%) | Tugas 2 (20%) | UAS (30%) | Nilai Akhir | CPMK1 (%) | CPMK2 (%)         | CPMK3 (%) | Status CPL |
|-----------------|-----|----------------|-----------------------|-----------|---------------|-----------|-------------|-----------|-------------------|-----------|------------|
| 1               |     |                |                       |           |               |           |             |           |                   |           |            |
| 2               |     |                |                       |           |               |           |             |           |                   |           |            |
| 3               |     |                |                       |           |               |           |             |           |                   |           |            |
| 4               |     |                |                       |           |               |           |             |           |                   |           |            |
| 5               |     |                |                       |           |               |           |             |           |                   |           |            |
| 10              |     |                |                       |           |               |           |             |           |                   |           |            |
| 15              |     |                |                       |           |               |           |             |           |                   |           |            |
| 20              |     |                |                       |           |               |           |             |           |                   |           |            |
| 25              |     |                |                       |           |               |           |             |           |                   |           |            |
| 30              |     |                |                       |           |               |           |             |           |                   |           |            |
| Ringkasan Kelas |     |                | Rata-rata Nilai Akhir |           | Target CPL    |           | 75%         |           | % Mahasiswa >= 75 |           |            |
| CPMK1-3         |     |                |                       |           |               |           |             |           |                   |           |            |

## Sheet 13: Rekapitulasi Ketercapaian CPL Program Studi

TEMPLATE REKAPITULASI KETERCAPAIAN CPL PROGRAM STUDI SISTEKIN 2026  
(Dok. 008) Formula:  $CPL\_Attainment(CPLx) = \text{rata-rata bobot nilai seluruh MK}$

*pengukur level M untuk CPLx. Target minimal dapat disesuaikan prodi; standar dasar nasional IKU 7 >= 50%.*

| KODE CPL | FOKUS CPL                              | JML MK PENDUKUNG | MK PENGUKUR LEVEL M (UTAMA)                                                                            | TARGET (%) | CAPAIAN AKTUAL (%) |
|----------|----------------------------------------|------------------|--------------------------------------------------------------------------------------------------------|------------|--------------------|
| S1       | Sikap & etika profesional digital      | 9                | FST-610<br>Capstone,<br>FST-612<br>PKL, MKU-507 KPM,<br>FST-714<br>Skripsi                             | 75         |                    |
| KU1      | Pemikiran logis, kritis & inovatif     | 12               | FST-610<br>Capstone,<br>FST-611<br>Metpen,<br>FST-612<br>PKL, FST-613<br>Sempro,<br>FST-714<br>Skripsi | 70         |                    |
| KU2      | Komunikasi efektif & networking        | 11               | FST-610<br>Capstone,<br>FST-611<br>Metpen,<br>MKU-507<br>KPM, FST-714<br>Skripsi                       | 70         |                    |
| KU3      | Pengambilan keputusan & manajemen diri | 6                | FST-610<br>Capstone,<br>FST-611                                                                        | 75         |                    |

| KODE CPL | FOKUS CPL                            | JML MK PENDUKUNG | MK PENGUKUR LEVEL M (UTAMA) | TARGET (%) | CAPAIAN AKTUAL (%) |
|----------|--------------------------------------|------------------|-----------------------------|------------|--------------------|
| P1       | Fondasi sains & matematika komputasi | 9                | Metpen, FST-612             | 65         |                    |
|          |                                      |                  | PKL, FST-613                |            |                    |
|          |                                      |                  | Sempro, FST-714             |            |                    |
| P2       | Sistem informasi & metodologi SI     | 15               | Skripsi                     | 70         |                    |
|          |                                      |                  | STI-205                     |            |                    |
|          |                                      |                  | Aljabar Linear, FST-408     |            |                    |
| P3       | Infrastruktur TI & keamanan          | 12               | Statistika, FST-714         | 70         |                    |
|          |                                      |                  | Skripsi                     |            |                    |
|          |                                      |                  | STI-413                     |            |                    |
|          |                                      |                  | Machine Learning, STI-624   |            |                    |
|          |                                      |                  | Integrasi AI, STI-520       |            |                    |
|          |                                      |                  | Data Mining, FST-714        |            |                    |
|          |                                      |                  | Skripsi                     |            |                    |
|          |                                      |                  | STI-521                     |            |                    |
|          |                                      |                  | IoT, STI-625                |            |                    |
|          |                                      |                  | Smart City, STI-519         |            |                    |
|          |                                      |                  | Keamanan Lanjut,            |            |                    |
|          |                                      |                  |                             |            |                    |

| KODE CPL | FOKUS CPL                                | JML MK PENDUKUNG | MK PENGUKUR LEVEL M (UTAMA)                                   | TARGET (%) | CAPAIAN AKTUAL (%) |
|----------|------------------------------------------|------------------|---------------------------------------------------------------|------------|--------------------|
| P4       | Platform digital & pengembangan aplikasi | 12               | FST-714 Skripsi                                               | 70         |                    |
|          |                                          |                  | STI-522 Mobile, STI-627 Platform Eng, FST-714 Skripsi         |            |                    |
| KK1      | AI/ML integration ke SI bisnis           | 11               | STI-626 Deep Learning, STI-624 Integrasi AI, FST-610 Capstone | 75         |                    |
|          |                                          |                  | STI-626 Deep Learning, STI-520 Data Mining, STA-701 MLOps     |            |                    |
| KK2      | Data engineering & ML pipeline           | 7                |                                                               | 75         |                    |
| KK3      | Cloud, IoT & infrastruktur terintegrasi  | 12               | STI-521 IoT, STI-625 Smart City, STB-601 Cloud DevOps,        | 75         |                    |

| KODE<br>CPL | FOKUS CPL                                  | JML MK<br>PENDUKUNG | MK<br>PENGUKUR<br>LEVEL M<br>(UTAMA)                                                                            | TARGET<br>(%) | CAPAIAN<br>AKTUAL<br>(%) |
|-------------|--------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------|---------------|--------------------------|
| KK4         | Audit, GRC & tata<br>kelola TI             | 7                   | FST-714<br>Skripsi                                                                                              | 70            |                          |
|             |                                            |                     | STI-519<br>Keamanan<br>Lanjut,<br>STB-701<br>COBIT,<br>FST-612<br>PKL                                           |               |                          |
|             |                                            |                     | STI-308<br>UI/UX, STI-<br>522<br>Mobile,<br>STI-627<br>Platform<br>Eng, STC-<br>501 UXR,<br>FST-610<br>Capstone |               |                          |
| KK5         | Platform digital<br>skalabel & UI/UX       | 14                  |                                                                                                                 | 75            |                          |
| KK6         | Manajemen proyek<br>&<br>technopreneurship | 8                   | STI-523<br>Manpro,<br>STI-728<br>Startup,<br>STC-703<br>Digital<br>Product<br>Mgmt                              | 75            |                          |

## Sheet 14: Ringkasan Implementasi OBE

*RINGKASAN IMPLEMENTASI MODUL OBE - S1 SISTEM DAN TEKNOLOGI INFORMASI  
2026*

| INDIKATOR UTAMA                          | NILAI   | KODE CPL | JML MK PENDUKUNG | JML PL TERKAIT |
|------------------------------------------|---------|----------|------------------|----------------|
| Profil Lulusan (PL)                      | 4       | S1       | 9                | 4              |
| Program Educational Objectives (PEO)     | 3       | KU1      | 12               | 4              |
| CPL (Capaian Pembelajaran Lulusan)       | 14      | KU2      | 11               | 4              |
| MK Wajib (Paket Ditempuh)                | 49 MK   | KU3      | 6                | 4              |
| MK Elektif Ditempuh (1 dari 3 Peminatan) | 6 MK    | P1       | 9                | 4              |
| Total MK Paket Ditempuh                  | 55 MK   | P2       | 15               | 2              |
| Total SKS Paket Ditempuh (Syarat Lulus)  | 146 SKS | P3       | 12               | 1              |
| Total MK Portofolio Ditawarkan           | 67 MK   | P4       | 12               | 3              |
| Total SKS Portofolio Ditawarkan          | 182 SKS | KK1      | 11               | 1              |
| Peminatan Spesialisasi                   | 3       | KK2      | 7                | 1              |
| Semester                                 | 8       | KK3      | 12               | 1              |

| INDIKATOR UTAMA                                | NILAI      | KODE CPL | JML MK PENDUKUNG | JML PL TERKAIT |
|------------------------------------------------|------------|----------|------------------|----------------|
| Syarat Lulus Minimal<br>(Permendikbud 53/2023) | 144<br>SKS | KK4      | 7                | 1              |
|                                                |            | KK5      | 14               | 1              |
|                                                |            | KK6      | 8                | 1              |

| NO | CATATAN VALIDASI AKADEMIK                                                                                                                                                                                                                                                                                 |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | 14 CPL (S1, KU1-3, P1-4, KK1-6) disusun dari 4 PL sesuai SN-Dikti + APTIKOM IS2020/IT2017; terverifikasi penuh pada Dok. 002, 003, dan 004.                                                                                                                                                               |
| 2  | Pemetaan CPL-MK dan level I/R/M didasarkan Dok. 004; konsistensi terhadap CPMK dan Sub-CPMK telah divalidasi pada Dok. 007.                                                                                                                                                                               |
| 3  | 21 BK IS2020 + 15 BK IT2017 dipetakan ke CPL dan MK secara penuh (Dok. 003). Termasuk BK-IS20 (Ethics), BK-IS21 (Internship), dan BK-IT15 (Global Professional Practice) sebagai pemenuhan adopsi BK kompetensi utama wajib. Kepatuhan IKU 7 (Case Method + PjBL $\geq 60\%$ ) terverifikasi di Dok. 008. |
| 4  | Sheet 9-12 adalah template implementasi asesmen. CPL program sah dihitung setelah data nilai aktual semester berjalan terintegrasi ke SIAKAD.                                                                                                                                                             |
| 5  | Sistem asesmen OBE lengkap (Formula CPL Attainment, 4 Rubrik Analitik Master, Siklus PPEPP/CQI, Radar SKPI) tersedia pada Dok. 008 dan 009.                                                                                                                                                               |
| 6  | STI-418 (Dasar Keamanan Informasi) = 2 SKS Teori (Sem 4). STI-625 (Smart City & Pemerintahan Digital) = 2 SKS Teori Non-Praktikum (Sem 6).                                                                                                                                                                |
| 7  | MKU-406 (Agama II) dan MKU-508 (Kewirausahaan II) = 0 SKS kebijakan UWG; tercatat di transkrip namun tidak dihitung dalam beban SKS kelulusan.                                                                                                                                                            |



*Dokumen ini merupakan Single Source of Truth Kurikulum OBE SISTEKIN 2026.  
Disusun sesuai Permendikbudristek No. 53/2023, Panduan OBE APTIKOM IS2020/IT2017,  
dan Standar LAMINFOKOM.*

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**Dokumen Resmi Kurikulum KPT–OBE SISTEKIN 2026**

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